

ActInf ArtStream 004.1 ~ Karin Kukkonen: "Literature as Uncertainty Practice"

ArtStream | Karin Kukkonen | 1:25:50

hello and welcome it's December 10th
2024 we're in active inference art
stream number 4.1 with Karin kinan and
federo and Bruno guests today we'll be
discussing literature as uncertainty
practice creative encounters with the
world there will be a presentation
followed by some comments and discussion
so thank you all for joining and to you
Ken for the
presentation thank you thank you Daniel
for inviting me um I'm speaking here
from the Norwegian Institute in Rome um
and I'm very much looking forward to the
responses from Bruno and Federico and um
whatever questions may come up uh in the
chat later
on I'm going to talk about predictive
Proccing in relation to
literature from the one side looking at
ways in
which we can use the framework of
predictive processing in order to make
sense of what literature is and what it
does um but also going the other way
thinking about what elements of
predictive processing literature might
um pose a challenge for or um provide a
very interesting case study to perhaps
provoke further discussion
a lot of what I'm going to say is based
on my book probability designs which
came out with Oxford University bre in
2020 um I'm going to introduce the
concept um I'm going to relate it to the

concept of form which is a concept from
that is much used in literary studies um
I'm going to raise the question of what
makes literature different from other
art forms and I'm going to link that to
a very particular kind of cognitive work
that literally form the long form
written of um a novel enables um and
then in a final move I'm going to talk
about the ways in which literature works
as an uncertainty practice in the world
and what this has to do with um the
general experience of
contingency so let's start with the
probability
designs um and being a literary scholar
I'll start with a textual example this
is from Virginia W's novel
Orlando um which is a novel where the
main character starts as a man in the
Renaissance then suddenly turns into a
woman in the 18th century and Carries On
as a woman until um about the 20th
century at this point in the book we're
still in the Renaissance young Orlando
has decided to become a poet until one
day he looks out of the window and see
he sees the splendid natural Green in
the garden after that
and here I quote Virginia wol of course
he could write no more green in nature
is one thing Green in literature another
nature and Ice seem to have a mutual
antipathy bring them together and they
tear each other to Pieces the Shade of
Green Orlando now saw spoiled his rhyme
and split his
meter so what we have here on a very

basic level is of course a prediction
era he wanted to become a poet um he
looks outside of know he sees something
surprising and he changes his plan in
terms of a narrative such prediction
errors would be plot events here
something happens that changes
fundamentally the trajectory um of the
further course of the narrative orlando
leaves his desks and his plans of
becoming a writer and what is likely to
happen next in the novel has just
changed these prediction errors are of
course um modulated with Precision with
our sense of how just how good that
information is just how reliable it is
and in literary text that is quite often
tied to the character's
perspective um so it's Orlando who sees
that green is part of his perception um
and it's very often also tied to the
embodied experience of the characters
and the way in which they're positioned
in the fictional World in particular in
the long form of the novel and this has
been discussed in literary studies for a
very long time at least since M B's idea
that the novel is a polyphonic genre
genre of many voices you usually have
multiple different perspectives on the
Character level but also on the level of
the narrator who can be more or less
reliable so that's very much um part of
a standard uh setup for the novel to
have these multiple different um
perspectives but it's not just the
spatial position
um that gives um Precision in a literary

text it is also on the level of temporality so if we take the last um sentence it has a particular Rhythm um that changes Midway through the sentence so the Shade of Green Orlando is I am then you have the now saw where the Rhythm sort of changes and that's of course exactly the moment of the prediction error spoils his rhyme and split his meter the second half of the sentence moves into iambic foot um wolf I think very consciously um replaces um the sentence in order to um make that um prediction error not just something that we rationalize but really something that you experience as a reader um on a quite embodied level so also the pacing of the pros the pacing of the language that we read um works as precision and guides readers attention um these features that wolf chooses to modulate Precision here the different meters um of verse are of course Very Much tied to the period in which Orlando is at this point in the Noel Renaissance poetry um he's in the Renaissance where poetry is one of the dominant modes of um literary expression and that takes me to the third level of the probability design the Precision expectations so expectations about where we likely to find um good um Precision the kind of parameters um within which we direct our attention so um the setting in the Renaissance the fact that he writes um poetry um goes together with the choice um of the meter that here and both I think

quite systematically develops that as she moves through the different periods so when we're in the Victorian period we read that sentences swelled adjectives multiplied lyrics became epics and little Trifles that have been sentence essays a sentence long we now encyclopedias in 10 or 20 volumes um you don't need to have read Dickens to know that Victorian novels are long um and wolf actually here even in Acts she's not just saying that Pros um has a very long binded format long binded syntax in Victorian novels she actually performs that um here as well again drawing attention to the framework within which we're moving so in literary texts quite often um the genre or historically specific modes of writing provide Precision expectations for readers and indicate where it is worth paying attention to while reading so the probability design then brings together all three levels the level of the plot and the prediction errors the level of perspective and pacing and um precision and the level of different styles of writing different kinds of intertextual references on the level of precision expectations um it's an Artful configuration of words that provides a design sensory flow and reshapes readers generative models for the duration of reading and then Design sensory flow I think sits between the sort of everyday setting where of course we have a sensory flow but it's not um designed

it's not ordered for us and cases of um
for example Dreaming or mind wandering
where we we're moving through the
imagination um as Andy Clark says in
serving uncertainty such systems of
dreaming and mind wandering command
generative model capable of
reconstructing the sensory signal using
knowledge of interacting forces in the
world that process of reconstruction
tuned and deployed in a presence of the
sensory signal paves the way for
processes of outright construction able
to perform able to perform and develop
in the absence of the usual sensory flow
so in the case of reading we have an
absence of the usual sensory flow but
that doesn't have to be reconstructed
completely on the basis
um of what we already have what we
already know instead it is reconstructed
on the basis of this design sensory flow
of the liter text that provides us with
um prediction errors in the first place
but then as I have shown um a whole set
of different elements that also model um
how we inference on the level of
precision and precision
expectation um what I talk about in
terms of probability
designs arguably can also be found in
other kinds of artworks and um for
example in the case of painting the
composition would be a kind of
probability design um and for that case
the work of Co bment um is very
interesting because they talk about the
ways in which the composition of a

painting has the capacity to create particular aesthetic emotions um and that is tied to a faster than predicted rate of prediction error reduction meaning um that you see a painting and you're quite startled it's not easy to make out what it is that you'll see but then because it is designed because the painting has a composition you can actually make sense of um what you're seeing in front of you at a much faster rate than you would predict at a much faster rate than what you would be able to do um in a real world setting and that creates a positive aesthetic emotion even if the painting perhaps depicts um something that would evoke quite negative emotions and I think something like that is certainly also true for literary probability designs that they have a way of um making you navigate um that probability design at a faster than predicted rate of prediction error reduction um but I would also say that there are quite significant differences between um paintings and literary texts just because of the way in which um they're made the way the things of which they consist in Reading literature um you usually have an engagement that extends over much longer periods of time than looking at a painting when you read a novel you spent probably about a week with it reading a couple of chapters here and a couple of chapters there so there is a way in which your everyday life and your reading process are folded

into one another as you go through um
reading that long form um fictional
text um reading a novel also asks you to
engage with visual stimulus but that
visual stimulus is a lot poorer um than
would you have in images it's just black
letters on a white
page so there are um quite important
differences that I think is important uh
to keep in
mind both um literary texts as well as
other art forms can create a feeling of
presence um in readers and that's um
based on work that I've done together
with Federico and colleagues looking at
Presence flow and narrative
absorption where we distinguish between
um motor intentions um that would be you
handling the book and turning the pages
proximal intentions
um where you use language to act or give
sense reactions of a character so you
experience um embodied resonances around
a character's movement for example while
reading that generates um core presence
and then there also distal intentions um
where you use the probability design to
make accurate predictions of what a
character will think or feel so they
develop a kind of strategy for getting
better off predicting um what that
character um will do um and that creates
extended presence so it has a kind of
second track or a meta track um to um
the inferencing process and I think for
the literary probability design that
level of the extended presence is where
most of the really interesting thing

happens and I think that is to do with
the way in which literally form or
probability design can move Beyond myis
can move Beyond just
representing um something that is there
I'll give you an example from a mid 18th
century short story called the noble man
where a young woman is escaping from the
shadow of her ancestors and she does
that by taking the portraits of the
ancestors of the wall and using them as
planks to run across the mold
um around the
shadow Julie jumps lightly on the face
of one of her ancestors as she leaves
the shadow she jumps on one of the
portraits um that face of her ancestors
renss and snaps beneath her
feet um there is of course here a highly
embodied language that invites embodied
residences and readers but at the same
time um that design sensory flow also
goes beyond the remit of real world
sensory motor
contingencies um I mean I'm I'm not sure
it's possible to actually walk across uh
canvas on water and not sink or I'm not
sure how fast you need to run for that
um exercise to work out that however is
usually not a question um that we ask
ourselves when we read that and that is
I think because on the level of the
probability design um all of that hangs
together all of that has its logic um it
is of course was very funny very ironic
that she uses exactly the sort of um the
elements indicating the authority of her
ancestors the portraits um for her

getaway so on the logic of the probability design um all of that hangs together all of that makes sense um and that enables um the literary text even though it draws very much on embodied resonances um to move away from um real world sensory motor contingencies and to reshape the inferences that are available to us moreover um Beyond these non-mimetic um moments literary texts very often invite what Pulo and colleagues have called epistemic active inferences that is you want to make a change in the world in order to see um what provides better information and in the case of epistemic active influences you would do that in your head you would scope for information by manipulating um the environment mentally readers I think constantly do that you're constantly constructing alternative scenarios along the lines of oh if she hadn't done that or um that character really should have come in time um Etc um you constantly project alternative outcomes sometimes texts even give you prompts for that but these epistemic active differences that forward scoping while reading um is a fundamental part of the reading process and I think it is most prominent or most noticeable when you read the book again that you've read several years ago um because it then quite often turns into a very different um experience the kind of sense that you make of it the meaning that it has for you tends to be very um

different just to give um a banal example reading anakaren before and after a divorce I expect would have a very different kind of um experience that goes with it and I think that has to do with the fact that yes we're reading the same text yes the plot events do not change um but the small uh inferencing processes that we run as readers um these epidemic active entrances are very much to do with the online present leing process and their fundament to the kind of meaning that we configure out of reading a novel and that changes of course between um different iterations of reading the text a second element um that I think is quite specific to um a literary probability design is the way in which novels allow you to explore inner states and of course the novel has um traditionally been considered as the cultural technology for the exploration of IM States and I think there is a predictive processing story um to that um insight and it is to do with what Andy Clark in his uh 2013 BBS piece called the dark room problem um where he poses the problem of um well if we want to minimize prediction errors why do we not um just close the door shut lines and of the lights um one of the responses or the possible solutions that have been put forward for this dark room Problem by pulo and colleagues is to say well actually there is an inceptive and an exceptive aspect to this we keep our

interception very stable so we don't want to have um a heart that beats very F we don't to be out of breath we don't want our tummies to be upset and if we keep that stable um um then we can be very adventurous exploring uh exception exploring um the outside world um and I would suggest that in the case of reading actually the opposite is the case we keep extra reception very stable um we don't want to be distracted we um focus on this very poor stimulus of just words on the page um keeping whatever attracts our extra reception um quite static

and that then in turn allows us to be quite adventurous when it comes to exploring our inates the ways in which we respond to what we've read um very much of course emotional responses and in the work of Lisa Felman Garrett interception has of course been um tied within a predictive processing framework to emotional

processes um but I think also things like um linking that to mind wandering Etc um would um fall within this exploration of in the states so the novel as a cultural technology exactly because it is um so stable when it comes to the stimulus allows for an exploration of interception and inner

STS there's the second aspect to that um which is perhaps even more speculative and that is to do with the way in which which um embodied resonances might um be affected by the fact that we're reading that as letters on the page so in the

study by Shill and all um they were interested in the ways in which you can get around sensory attenuation so the ways in which you can lift um the attenuation of the sensory feedback that what you don't want to have while you're moving um they use the of um the escalation of force so if someone pushes at you and you asked to push back to the same degree you will usually push harder than the pushing um that comes against you exactly because you don't have a very good sense of how hard you're actually pushing um that kind of um Force escalation can be apparently um avoid it if you use a kind of interface that redirects um what it is you're touring and arguably words on the page are a similar kind of interface so using um an interface for pushing that redirects our movement in this case would limit sensory attenuation you would have a better sense of what you're actually doing in an embodied um sense and perhaps also embodied resonances of words on the page might have a similar effect now like I said this is quite speculative but I think it would be interesting to see whether there is some kind of empirical reality to that and it would go together I think with this idea that literary texts as an exploration of inner states allow for more precise investigation of what's going on within us um then basic offline thinking would enable um the elements of

of reflection um can also be tied I think very much to the way in which prision is modulated in probability designs and here I suggest that literary texts have a way of constantly providing what I call Meta fictional nudges so nudges um that provide a meta perspective on the kind of immersive experience that you're having I take the term of the nudge from John Sutton's work on um Sports Performance where he talks about cricketers who use words for self-cueing um so that they can sculpt the ongoing flow of motion by saying things like battle or keep the battle something like that so you adjust what it is you're doing in the flow of the action and I think that's exactly what very often happens in literary texts I've given one example here from Francis Buffer's um the go Golden Hill where we're in the middle of a Fel Septimus disengaged with a WRA of Steel and lunch lower in SE Smith replied in C seek supreme but this is really useless and no more enables the reader to see the battle than if I shout numbers at you which indeed I appear to be doing so here you've got exactly that shift from a quite immersive description to these French numerals which of course indicate fencing decis uh fencing positions um first working as quite integrated quite subtle metafictional nudges and then being taken up by the narrator to a full-blown um meta fictional meta narrative

reflection and that's something that happens quite commonly in literary text and I think invites readers exactly for that kind of uh shift of levels also the interconnectivity of text that I've talked about in terms of that third order level is something that I think allows L text to do some quite particular kind of cognitive work and here my example is a novel called de composition angara where the main character is a serial killer who works on the basis of PRS In Search of Lost Time and the rule that he gives himself is that he takes up a volume of pro um identifies the characters say Madame um counts when she appears in a given part of the book so she would be character number 37 who appears in that part of the book then he goes and sits in the park and counts the passes by number 37 if it is a woman he will go after them and kill them and then go home and uh open an electronic copy of Prest Noel and remove that character from it so here we have a case of a very strong emphasis on um that in textual Dimension that um Precision expectation Dimension and that is highlighted even on the level um of the words that she uses for example after the description of the kill rule um the narrator says deushi with happiness joyfully um you're going to bed which is a riff on the first sentence of BR novel where it's the B Earth so early um I used to go to um so gar quite systematically uses not only the character repertoire of post um

but also um the language to establish that

interconnectivity um and I think that allows readers to reflect both on the rules and predictabilities of what's going on in that novel um but it also allows them to reflect on where they actually take their expectations from I mean is it the crime novel the serial killer narrative um is it the crime Society modern uh novel for which Cru um stands so as you're exploring Deton Library um this is something you can do um in the literary text exactly because it provides scaffolding for it because it ties it to different characters to titles of novels to ways of writing um Etc so that exploration of the mental library of different kind of precision expectations or sets of precision expectations and that is a term I take from PI the mental Library um is also I think something quite particularly suited for literary texts um because it allows a moment of stability and it makes these predictions are U sorry Precision expectations perceptible um by highlighting them and making them part of the story itself so to summarize briefly um what I think is quite particular about literally probability designs is the huge space they give to epistemic active inferences um it's the enabling of the exploration of inner states and I think that is in particular linked to this long form written text where you keep

coming back um to the same fictional
World while your everyday life um is
going

on um and thirdly that it allows you to
attend to Precision expectations this
third order level um of cognition which
in an everyday setting in the ongoing
flow of action um you cannot really
attend to quite as easily so the long
form written probability design of the
novel um enables you to do all of that
cognitive work um to an extent which is
hardly

imagin hardly imaginable in an everyday
context um I think all of that is tied
to form and probability designs um very
much corresponds to a general notion of
form and I take my understanding of form
from fed Schiller's book on the
aesthetic education of mankind he sees
form as a means to provide freedom for
the human mind and here of course the
epistemic act of entrances this roaming
of the mind um is one um element that's
quite important but that I think is made
possible by the scaffolding provided in
the literary text exactly through that
design sensory flow that keeps um these
movements um discernable and thinkable
that understanding of form as a space of
Freedom as a space where you can watch
yourself think um is quite different
from how form is generally understood um
certainly in literary studies Caroline
lavine's recent book um forms which has
um been very influential in literary
studies says well there are social forms
and there are also aesthetic forms so

social forms for example would be the layout of a classroom and the aesthetic form of the novel um can pick that up and integrate it for example in a novel like um J A where we have a classroom setting um and she makes a very interesting argument I think about the common durability between aesthetic and social forms but she also insists that forms as designed arrangements are deliberately crafted to impose order just like art objects um and I think is perhaps a little further than I'm willing to go because certainly probability designs provide patterns they provide scaffoldings um for us to give structure um to the experience of reading but I don't think they very strictly speaking impose order um for LaVine um the only way in which literally forms can create contingency can create a sense of that things things don't necessarily are the way um in which they appear to us now that there's always an alternative that um happens when you have different kinds of form collided but form in and of itself wouldn't give space to that kind of contingency exactly because it imposes order and I think the probability design idea allows you to rethink um the role of contingency in um literary forms uh and I've explored that in particular in my most recent project where I look at how authors create um probability designs how they

write their novels um and there I suggest that contingency not only plays an enormous role in the creative process itself but it is actually inbuilt in literally form that also deploy form um to integrate contingency into that text to always face um for alternative ways of interpreting things making that available to readers so form of course encompasses contingency it is a contingency trap as I suggest but it's the kind of trap that you see at the bottom where the mouse will be caught um but it can still run around and it gets stabilized enough for you to observe the kind of things it's actually doing um form is not the kind of contingency trap that you can see at the top that just kills the mouse

dead to give you an example of what one of these contingency traps um looks like um I'll briefly talk about italino castle of Ross

Destinies where in the very process of narration you have this very alator um chance-based composition process where tarot cards are placed on the table and the narrators develop their narrative from the card that they happen to draw and place on the table so on the level of narration we have a representation of contingency but much more interesting it also thematizes the way in which contingency is inbuilt in the form itself because in the first iteration of that narrative technique the Middle Point the middle spot of the tarot um cards remains empty and um this

is work by Natalie burkeman um who suggests that this is a case of the clean men um the swerve or deviation from a planned protocol um the swerve or Camen in terms of a probability design would be an underspecified element um that provokes further inferencing because the middle spot in um this tarot tapestry is kept empty you can always generate um new stories out of it and similarly literary texts always need to have some kind of flaw something that doesn't add quite up in order to um provide exactly the space um to explore contingency while reading so I think in the very design of a probability design um there are very often these moments of a break or a flaw that enable these Freer um formal patterns of inferencing and meaning making with that in mind I want to come to the last part of my talk where I address the issue of how literature works as an uncertainty practice uh in the real world um and here I draw on an article that I published last year where I also draw on the piece by Miller and colleagues um where they talk about the value of uncertainty in the context of algorithm um and the way in which algorithms um make volatile environments more manageable um they talk about personalization algorithms so like when you order a book and you get five recommendations on the basis of what

you've ordered before these kinds of mechanisms that make the world more predictable more um according to how you've behaved in the past but they also have a very interesting example of someone who programs an app for his phone um that randomizes his food deliveries so he never knows um what he gets so you can use these Technologies to go in both

directions um and make sense of um the volatile environments that we live in they briefly touch on literature in that article and they say that it's a safe space or a safe kind of environment for

exploration which in a way I think goes together with a lot of things that have been said about literature in a sort of larger um context where this is a space where your actions wouldn't have consequences where the killer from um gar novel doesn't come out of the novel to get you but of course you can also not save any of the characters in the novel so it is a space um of its own where you don't um need to fear consequences of actions so in that sense of course it is safe um but I think it is a lot more than just a curated space for Thought

experiments um I think it is a lot more than that exactly because of the ways in which reading a literary text has a way of tying itself um into your everyday experience as you do that over multiple weeks because of the ways in which it draws attention to your reactions um to

what you're reading so for example
inos Madame or the person identified as
Madame by the killer actually gets away
and that goes very much against um what
the whole novel was set up to do um and
I think at that point you would perhaps
ask yourself why perhaps you Shar the
disappointment of the killer then you
might ask yourself why you do that um or
you don't do that and then you're
wondering where the novel is going to go
um so the way in which the liter text um
is set up making things uncertain um or
contingent is often also an invitation
to reflect on how you react to that um
yourself both with the exploration of
inate but also in tandem with that
metafictional nudges that I was talking
about earlier
on and I think that even applies for the
question of um Precision expectations
where the way in which this is set up in
G to stay with that example from the
moment that Madame Loop gets away we've
got continually new iterations of
different character kills in different
narrative modes um there even is an
evocation of an ego shooter um where the
killer based basically depopulates the
entire world of In Search of Lost times
uh In Search of Lost time and that
variation of narrative speed um I think
is a way of drawing attention to um the
different regimes of attention that we
go through in that novel The ego shoter
going very fast very quickly and of
course proves Noel because of the way in
which he writes and how long the thing

is um which grows very slowly so that
would allow for example for reflecting
on um the different kind of
accelerations or attention regimes um
that we operate in while reading that
Noel the herself actually in an
interview draws attention to the way in
which the experience of reading a book
is by no means a safe environment she
says it reorders the circulation of your
affects it manipulates your desires your
perceptions your efforts in ways that
are not predictable
so literature to conclude my
presentation and its probability designs
are actually quite strange think about
Julie running across the portraits of
her ancestors and not exactly safe
because theyve constantly forced you um
to reflect on what's going on to reflect
on how you think about yourself they
have the potential to at least while the
reading process is going on to reshape
the generative model with this design
sense Reflow and they make elements of
cognition that are usually invisible
available to
reflection so they enable creative
contingent ways to rethink yourself and
the world and I think that's as much as
we can ask of any kind of cultural
technology thank
you thank you very much Karen was really
fascinating account of um how literature
can be seen through the lens of of
predictive processing and an active
inference and I think I would like to um
because you dived really directly into

how to do this how to use this framework
to to think about literature and how
readers engage with the literature but
maybe other question that can be
interesting for for a broader audience
and this is a um so what do you think
that this perspective is offering um
that is new or that is more effective in
terms of um how we can better explain
what's going on in literature and when
we engage with literature in comparison
to um existing Frameworks or alternative
uh theories that deal with the with
similar um aspects of
literature um that is an enormous
question

um I think one of the reasons why I've
decided to spend quite a lot of time
with protective processing um to think
about literature is that it allows you
to look at multiple levels together so
it allows you to think about the plot
but also how that's connected to the
character and how they move through the
fictional world and then you can also
make links to um yeah very particular
kind of genre protocols for for example
or intertextual links um so I think
because of the the way in which it works
across different levels um and still the
flexibility that I see um in the model
um what I like about it in terms of
thinking about literature why I think is
effective is that it actually allows you
to think things together that
traditionally in liter studies we don't
think together so character and plot um
are usually two different

strs of things that you're looking
at yeah great answer and actually made
me think about

[Music]

another maybe issue that it's uh very
often uh debated or maybe not enough
debated sometimes in in literary
theories and is about the need or the
possibility actually to have a
comprehensive and Universal theory of of
narrative of fiction of literature and
by what what you've said and also this
is my impression um in other Reflections
that well it seems like that something
like um thetative processing framework and
active difference similar to how they've
been proposed in the cognitive Sciences
can aim at being this um overarching
theory for describing and explaining
different aspects of narrative and so my
question is do you think this desirable
and it's possible and how this can be um
can go together uh with the need uh that
has been expressed multiple times in the
humanities of having different
perspectives and different
um people with different theories that
can actually bring us to a deeper
knowledge of a
phenomenon

um again a very big question um

I

think with predictive processing you can
ask very different questions of a text
um and you're
not tied to so if we're thinking about
universals um that very often comes with
a category or a set of categories so I

don't know four different character types and then you've got everything in compass um so it very often comes with a top

down um imposition of boxes so you've got the hero and the helper and just to take a very basic example of Pop's fairy tale studies the

structuralist

morphology um what I like about predictive processing is that you can do that bottom up um you don't have to look for a particular character category in a text

um so you can with that account for um any number of narratives without um presupposing heart distinctions between types or structural boundaries instead you can trace how the text or the novel itself establishes um those categories or keeps them open I mean that I think that is a lot more common in in literary novels that the and um the hero um are not very easily to be

distinguished um so I think one of the attractions is this bottom up element to break processing well that it allows you to do bottom up analysis that can all be analyzed with the same analytical vocabulary yeah thank you yeah I like the the the perspective and I'd like to to ask you last question before uh letting the the floor to to Bruno and I know that you in your work um and it's an example of how you uh presented this today you look at text from different epochs right not just contemporary

material like to look at how this Theory
can work with the with text from from
the past so I always wondering how um
this can work well uh
this way of of looking at the text and
thinking about probability so kind of
cognitive process that readers are doing
or um hypothetical readers are doing and
how do you in your experience in your
analysis of the of several literary
works that you've done how well do you
think this is applicable to text from
the past and uh or where do you see the
limitations
are um so I think it is
very well applicable um in fact I got
into predictive processing because one
of the central categories of early
modern literary theory is
probability um and I was working
on 17th 18th century literature looking
at ways in which how they thought about
what was happening when people read how
that corresponds to current strands in
cognitive science our philosophy of mind
so um for probability I actually started
um from the early modern period where it
was very important that the text was
probable um and went from there to see
well actually predictive processing as a
theoretical model Works quite well to
understand how these um probability
calul
calculus um actually work so
interestingly enough i i p the other way
around
um and certainly in the early modern
period the way you thought about um many

literary works not all of them was you
know um not is it true or is it false
but is it probable is it well enough
designed does it have a good enough plot
to hold together so that you believe it
whether it's true or not um so I think
predictive

processing works very well

um for I I mean one would have of course
to put it to the test and one would have
to apply it to Medieval texts or ancient
texts as well um but certainly for the
periods that I know which is mostly the
early modern um period it it works quite
well um it would be interesting to see
what would be the the sort of limit
cases where um the idea that you've got
an ongoing um calculus that the
trajectory of the narrative is Con
updated um whether it wouldn't work I
think that would be very interesting to
see what kind of narrative that
was thank

you okay so I guess I'll just move on
thank you uh for that lovely
presentation and and for sharing your
book it's been quite an interesting read
and there really um changed a few things
how I think a few things in my field
um I just want to say that I come from a
very different field um anthropology and
archaeology but I think that a lot of
what you say actually there's some
parallels between the work that we've
been doing um just to
specify uh that worked mostly with the
breakage of form and I I'll ask a few
questions about this

um uh

particularly uh in terms of craft work
so a lot of work I do is on Craft work
and there's an interesting point you
rais in in in the book about and and
also in your presentation about how
reading what what makes literature so
exceptional in a sense what makes it
different and I'm wondering if you find
any parallels between which of the
processes of probability design with uh
with craft work with actual work with
with materials um yeah I'll start with
that

um so so you're thinking of the actual
Act of

making a pot for um yeah sure

yeah um I think there

are similarities certainly I think I I I

have I have to admit I have no idea how
you make a pop no worries um but I I

assume it is also a more long form

engagement it's something that unfolds

in the flow of time and um where the

material that you're working with gives

you different kinds of resistance um and

you have different strategies of of

working with it or working around that

resistance so I think in in that way

it's probably similar to the reading

process that it is something that

unfolds over time and where you get a

sense of what is likely you know going

to work um what kind of approach you're

like

to yeah it's is likely

going to help you make sense of what it

is that you're reading um so I think in

that sense um there are a lot of similarities there um I think one of the differences is of course that reading requires you to work with language whereas the material um of the pot is not strictly speaking symbolic um I'm very happy to talk about the symbolism of a material form um but it's not uh first and foremost first and foremost it's matter that you shave um so I think there is a difference with um how you handle and manipulate that in your head um from what you do as a reader yeah I'm glad that you say that it's been something been debated a lot in our fields in terms of there's a turn around trying to make a looking at materials as as as if you were reading a text a lot of complication arose which you just synthesized very well I think but um I think I'll I'll shift to the the next Point really is is in terms of form I'm quite interesting this point because you seem to suggest that form is something emergent which I think is partly fueled by this U by the active inference framework um and if I understood correctly would be one would be in this sense um uh the design sensory flow yeah um and there's this tendency we have of trying to kind of petrify it maintain it make it durable um and uh but at the same time we're facing challenges right in the text from when so I would be interested

in looking at if there is form in the
sense what would be the breakage of the
form so what what would be something
that challenges the form when when when
when is reading or in terms of
probability

design um I think there are multiple
ways in which you can do that
um but before I start talking about that
I I think I mean I would be interested
in actually hearing your take on that
because it it seems to me again as
someone who's not very well informed
when it comes to craft that if you do
have a material object I mean you speak
about fossil

and I mean these are stable
configurations M

um that of course you can't perceive the
whole pot at a time but you can get an
outline of it so if you or if you look
at the painting you know you can do a
composition sketch um that puts together
all the dominant lines and the outlines
um so that you can see where the
triangles are on the center of gravity
or um the the the vanishing point Etc so
it seems to me that in the case of
visual art and and also material objects
there is a way in which the form is much
more immediately

perceptible whereas um in a literary
text and especially in a long form text
like the Noel

um it's really only something
that yeah it's it's somewhere there in
the text but I think as soon as as you
try to draw a diagram uh as soon as you

try to you know do these plot lines or
whatever people have
attempted um it sort of doesn't quite
capture
it I think you do have an intuitive
sense of um what kind of shape the whole
thing has and I think there are various
strategies when you start a Noel um of
projecting what kind of ending it's
likely going to have um evoking General
protocols for example is establishing a
setting that will come up again at the
end Etc so there ways of signaling
formal coherence in a text um but I
would be hardpressed to imagine how that
would have the same kind
of perceptual Unity I think that I'm
looking at AAS for example
has um still I spoke of course about um
cracks and flaw
in um the probability
design um and I think those are um
exactly these moments that are heavily
um
underdetermined um that like the the
character getting away in in the Cru
killer novel when everything has been
set up to say well this is not going to
happen this is completely impossible and
it it's also never explained properly um
one example that immediately I think
comes to mind or another example would
be sort of what happens between
chapters when characters at the end of
the chapter Clos the door and then the
new morning starts the next chapter and
you you sort of know what happened in
the night but it's not made explicit so

I think there are various points um in the narrative or in the novel um where you can position those cracks and and those flaws um and I think some of them are to do with the very material layout like chapter endings um but mostly I think it will be on the level of how the narrative is constructed does that answer your question yeah I think it's it's true is it must more of a it's much more fluid this idea of of of of form really so it's hard to disentangle the this what would be the exact let's say not the exact opposite but more or less the case about how how form is really dissolved if to phrase it that way um I think there's a point I I want to ask you about um in terms of because you mentioned like uh plot events or these things that that did not happen a lot in and maybe I'm referring to exclusively on on your book in that case all about the this narrated and um and I'm Wonder wondering you you raised a really interesting quote that um I think I have it written here I can say directly yeah the unfulfilled possibilities or to put it more dramatically the hypothesis that die with each transformation in a plot event stay with the probability design and I actually think this is quite a crucial remark in that sense to understand how these let's say interruptions that are flowing through the the text actually come to shape the the this form as you

as you're describing um so my question is are they would it be is there a selective process into how some hypothesis might die out why others stay with the POs sorry probity design is that um maybe so bit confusing the question but um I think I understand it um so I think that is very much to do with the individual reader um and how you move through the text so if we think of a hypothesis that dies um I don't know that an now St with a husband you know which is possible at multiple points in the book um it is a very dead um hypothesis though but of course you could read the novel and at every point you know imagine how you know her life could have been different it may even have been better um I think that depends on your take on Roni um but I think it is very much to do with um exactly the kind of epistemic active inferences that become dominant across the reading process and it's very much to do I think also with the kind of emotional investment that might have as a reader because you feel sympathy with a particular character or because you think oh this character is a lot more interesting than I actually thought um often when you read a novel The Second Time Around and you start noticing the things that you haven't noticed um the first time around um so I think I don't think there is um I don't think it's predictable I think that is exactly which of these hypothesis

that stay with the probability design
even though they don't get realized I
think that very much depends on what
happens um during the reading process
what for whatever reason is attractive
for a reader to carry with him or her
through the
text okay I see your point um in terms
of this rereading I think maybe I'm
passing too much time so I'll leave this
as the last question before now um in
terms of the rereading uh process um and
these ideas that these processes that
are adding uncertainty let's say to the
to the narrative um
so would would that be reconfigured in
terms of these these hypothesis with the
rereading
um are some of these taken or are they
some of them just left to die
eventually because of this constant
reevaluation or reiteration of of the of
of looking through the text and for the
reader I mean
yeah yeah I think most of the you know
possible Alternatives will never cross
your mind um because there are just too
many of them um so I think the the one
if you do carry an alternative
hypothesis with you um it will be one
that for one reason or another you have
an emotional investment and all you find
particularly interesting at given point
um so I think it it
leaves potentially an infinite range of
making epistemic active influences but
you won't have the time or the interest
or the energy to make all of them um I

think it will be a very few of those okay see Point um yeah sorry I said it was the last one so I'll keep my promise yeah maybe I can add one small things it's a comment on your previous question and to C answer because we touched upon this in the article me and K have written together and then you mentioned today this distinction between core presence and um extensive presence right so the difference between the proximal intention and the distant intention so you can make predictions and and then accept the errors coming through out of them as long as they don't disrupt too much the the goal that you have the the the distant intentions that you have and they are driven by situational factors of by Jean convention let's say that you're reading a detective novel if you keep making predictions about who the murderer is and the murder gets revealed on page three that would be a to an abrupt disruption for you to accept that and then wanting to keep engaging with that kind of narrative and and and so you you ask if there's a trade-off or some uh things that um are not included in the probability design or get discarded in the writing process because they're not they don't work well in the overall design so maybe this is an example of something that could in some cases if you're really not if you're writing something that is subverting the Jean convention and not really um putting the emphasis on who's the um the

murder but wanting to play with the the
Gen of the detective story but this this
could work but in most cases for most
readers that want a detective novel um
that wouldn't

work I see um that's is quite
interesting because uh what I see is all
these what I find interesting because
for us when we're working with craft
work on certainty seems to be of the
external world you know it comes in it
everything's messy um the things Decay
they wither they break down so you need
to adjust whereas here find it
interesting because a lot of this uh
uncertainty comes from the actual design
of the of the of the of the text as as
as it's been uh described and so in that
sense

um is there an element to this to this
design in the writing that one
can one can try to see these these um
let's say these hypotheses that uh that
were never written

do you find traces of these in the
design for
example when when when the text is
designed do you find traces of these
doubts let's say of the of the
writer when when they were do you know
what I

mean something that might be external to
something

that was intended but is can can one
highlight these in the text about about
uh the moments of doubt the author has
when they're writing it the reason why
I'm saying that is that is there

uncertainty that is created
inadvertently when it's when the text is
being written that's that's my point I
don't know if
it's yeah I think um that certainly
happens because I mean I talked about
how reading is a long-term process but
of course writing a novel is an even
longer
process um where if you look in into
manuscripts you can see authors you know
changing their minds um and that will
leave traces um as they keep copying and
editing um what they've written um and
again with a text that is as large as a
novel is I mean all of these are of
course meticulously edited but I think
there is space for you know the mess as
you call it um to come in from the
different manuscript um
stages um sometimes that even continues
after publication um one of the early
modern authors I've worked on Samuel
Richardson um he wrote a novel called
Clarissa where one of the protagonists
is an aristocrat and he thought he had
written him in a very unattractive way
and nobody would like him and of course
all the readers or many readers thought
you know he was the real hero of this
piece
um so there must have something gotten
in there um that he didn't so it's not a
question of um you have an intention and
then you write that there's always stuff
that gets in there I don't know drawing
on cultural Protocols of Rish
Aristocrats that have attraction um

without meaning um to do so and then of course I mean Richardson then spend you know various additions trying to edit out those elements that he didn't want in there in the first place so I think that that is a very common um phenomenon actually that you've got um a lot of mess um even in in a very well designed probability design okay see I can relate to that it's it's a very interesting question if I have time to also add the small thing Daniel is it fine yeah no because well some may even argue that so the uncertainty but not in the sense of of the mess right of the but in the sense already of of how uncertainty is defined by productive processing might be um the core uh quality of uh of literature that lasts long and actually so Ken presented the work BYO calino and he has written an essay um um I think the name was is what is a classic and and his definition of what is a classic is precisely this text that have something which is left open so a level of uncertainty and it can pertain to many of the different aspects that the carent touched upon that is is uncertain because of the overall design of of the text and and how it works and it's open also for future generations and that's why it becomes a classic right because this uncertainty gets transmitted and and the text gets transmitted because of this um leing uncertainty that appeals to to readers also of the future that

that Calino couldn't uh think of or any
any author of Classics couldn't think
of it's quite quite fascinating because
something that stems to mind this
is how

would these different scenarios of
uncertainty I mentioned uh something
that you Ken used a an example in her
book about these things that do not
happen really so
how how would those be scaffolded in the
text you are they is that um for the
reader to be able to follow through on
these

details does it make
sense are you saying it it should be
there there should be textual signals
that you can point to yeah yeah so how
would it be okay I'll give you an
example maybe I'm thinking too much
about

craft so what we see normally is that
when when we set out to learn about
let's say you mentioned Potter so I'll
throw in

Potter um set out to make a part and the
last thing in your mind you know that
you encounter problems the most thing
most of the things you want to learn
about is the techniques to make the
object right but the last thing in your
mind is what will all the that what you
will effectively learn eventually are
actually all the faults that will come
out throughout the whole process right
but you don't intend on learning this
and it's not explicitly taught many of
the cases okay so a lot of this

uncertainty comes from the exposure to these contingent situations yeah um and the problem here is because it's not explicitly taught there are ways in which regardless if it's not explicitly taught what I mean is there are ways in which somehow it's there's a sort of scaffolded learning in the process of something that's not explicitly stated and I'm wondering so for example you would say the first thing you learn when the fault appears is um is affecting what is the fault you know what what consequences would this have you know then the next stage you learn okay um what do you do in this situation and then at the last stage you would say you learn to anticipate it right so you learn to anticipate this uncertainty yeah and so I'm wondering if it's a similar and maybe I'm thinking too much about craft but I'm wondering is a similar process with learning all these um let's say situations where uncertainty is being fed into the narrative to to us to uh get the reader engaged let's say with the with the story so does that make sense a little bit better maybe I think it does though I'm beginning to wonder whether that might not be a point where craft is actually quite different from Reading because I think in the craft you want your strategy to anticipate um those flaws in in order to

not have them have
whereas I think when you're reading a
literary text you actually do
want that kind of opening um you want to
be
given that kind of Freedom um so I think
you have a similar process of developing
exactly that
um the capacity to anticipate you know
if if the Killer is revealed on page
three um if it's a standard crime novel
um you don't expect much more but you
also know that if someone I don't know
like go chrisy does that so maybe there
is then something else coming that Pro
provides a new way of integrating that
information in a probability design that
is interesting in a different way
because you know who the Killer is but
maybe there's a new question that you
hadn't even thought about
um so I think there is similar
development of let's call it expertise
um as you work with the material in that
case of reading a book um but I think
you
don't this anticipation is not to
prevent um the floor from coming about
it's we need to see okay so now I
anticipate the floor and I want to see
what it yields I want to see um what the
consequences are so I think there is a
difference there um and it may be to
do yeah I'm not entirely sure um what
element that would be um based on but
perhaps it is to do with um letters
being I mean they they don't need the
kind of structural Integrity that a pot

needs to hold together so maybe there is something on the very materiality of what we're working with that allows for um that kind of

looseness MH I get your point that's certainly true yeah sorry yeah

towards a a pause in the chapter a chapter break here thank you all for the work I think it the the book and the other writings have opened up many avenues for people to explore and I'll ask one question which is

art has always been cognitive how though has this cognitive turn in explicitly exploring the generation recognition and interpretation and Analysis of art how is that explicitly cognitive turn re-entering into art

itself um that is a very good question um I don't actually know um I mean there are obviously writers who have a great interest in various kinds of cognitive theories um so who read that because they're also very self-reflective and interested in you know how this works and so arguably there are writers that take up

um cognitive science Neuroscience um Etc I'm not familiar with anyone who's taken up predictive processing though i' be of course very interested in that um so if you're talking specifically about the the sort of creation then I think it depends very much on what writer read does that answer your question

yeah it's uh I mean it's an opening to see how this all happens but like the craft maker was already doing what they were

doing what does an explicitly cognitive lens and a unifying cognitive scientific approach to memory anticipation the layers of probability designs that you articulate how does explicit knowledge of these in contrast with an implicit tacit

craft Artisan approach to generating I mean

icting is still in play even if we were to take that into a sort of forensic literary cognitive

science approach so certain things are in common and yet

now increasingly aspects of the cognitive

elements maybe in the future people are reading with

augmentation and there's new accessory possibilities for

interoception while you're

reading or there's generated

texts not just recommended or Word of Mouth

dissemination but instantaneous generation of

designed probability design

the these are these

are shaping the structure of the form in the

media yeah I think what you raise about

the implicit and the explicit is very

interesting because on the one hand of

course it is an implicit process I mean

you you go and you do it um on the other
hand I do think that the this meta
dimension that I highlighted with the
nudges um it's quite important
especially for the case of literature
that there is a constant element of um
adjusting taking a second order
perspective um so you could very well
use um say an analysis of a Noel
appropriative processing where you
actually identify and tag the
elements um in order to think about what
it is that you've been either reading or
writing so I can see very well that um
that kind of analytical language can
provide um nudges or metap fictional
nudges um for readers or
writers
um

so I think there is a place to think
about predictive processing not only in
terms of the theory that may explain um
what is going on and how how it hangs
together but that could be used
practically um as
well perhaps the panelist and then Karin
any closing thoughts or how you take
this
forward perhaps Bruno then Federico if
you have any
comments well I think um I'll be
brief but uh I think there's lot of for
me personally is got me thinking a lot
about
um how
exactly the sequences of events are
structured in in my line of work and uh
how to look at these things a little bit

more fluidly um with all these interruptions maybe I think we tend to be a little bit more schematized with with the way we engage with materials and so I think this is a way of actually understanding the process as as it's as it's unfolding really um and uh yes I think the the second part has to do a lot with this idea of uncertainty and and contingencies which are seldom used in in in our field so I think it's been really inight in that in that regards to look at um how this is actually an inbuilt part of the design process and and uh people come with very different ways in so as to how to respond to these different um elements yeah sorry that's very very interesting perspective in see how um looking at archaeology and crafts can can have commonalities with the with and I I think um what I would like to see done more and also this what we've been doing when when writing this article together and thinking about okay this is it works with with narrative and and literature but then can we use this framework to see also what is specific about literature as as a text in comparison to to other media right because we live now in a society and in an era where literature liter text are not the primary medium through which people experience fictional words and

narrative so this I think because of how
it can work well across different
mediated experiences I think the active
INF framework can be a very um good way
to define what is specific to certain
medium or a certain genre even if you
want as a the specific type of using a
medium in in a certain way um in
comparison to how can tell stories uh
through a different media so I think
yeah I think this would the the the line
I would like to the line of research I
would like to read more
about thank you I think it's really
interesting how we're coming back to the
question of the medium um which has been
so much discussed for media that are not
literature um which has been discussed
for film and virtual reality Video Games
Etc but not as much for literature the
fact that this is a written text that we
spend so yeah this much time with um
which is it has an effect on it and
which is different from the craft
example that um Bruno brought um so I
think from from our discussion now that
is something I want to think more about
um the the the fact that this is a
medium in and of itself and that exactly
specificity that filo highlighted is
very much rooted in
that
meeting cool the kinds of things that
happen on the live stream
medium thank you though all this is
very awesome I hope people enjoy it too
so till next
time thank you very much thank you

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